secrete chitinases, lipases and proteases which can dissolve the cuticle. The hyphae enter the haemolymph and proliferate and colonise the entire insect and release blastospores. Insect death occurs due to nutrient depletion of the haemolymph or by toxaemia by secretion of toxic metabolites.



Figure 11.11: Picture of insect infected with *Beauveria bassiana*



Biopesticides registered in India:

1. *Bacillus thuringiensis var. israelensis*
2. *Bacillus thuringiensis var. kurstaki*
3. *Bacillus thuringiensis var. galleriae*
4. *Bacillus sphaericus*
5. *Trichoderma viridae*
6. *Trichoderma harzianum*
7. *Pseudomonas fluorescens*

11.3.3 Viral Biopesticides

Viral insecticides are pathogens that attack insects and other arthropods. Viral pesticides are used to control Lepidopteran larvae like *Helicoverpa, Spodoptera* sp on Cotton, Corn, Sorghum, tomatoes. Baculoviruses are the commonly used viral biopesticide. They are extremely small and are composed of double stranded DNA. The genus Baculoviruses contains 3 subgroups.

- Nuclear Polyhedrosis viruses (NPVs)
- Granulosis viruses (GVs)
- Non occluded viruses

Mode of action of NPV



The virus enters the insect body via ingestion by insects and infects the midgut cells by membrane fusion. The NPV uncoat within the nucleus of cells and pass through the intestinal epithelium (Figure 11.12) and establish a systemic infection of the haemocoel.

Symptoms

Discoloration (larvae turns brown or yellow)

- Decomposition or softening of larvae
- Lethargy
- Infected larvae hang upside down twigs
- Larvae become swollen with fluid containing virus and eventually die turning black in color.

Mass production of NPV

NPV are mass produced in laboratory using suitable larval hosts. The fifth stage larvae are fed with food infected with NPV. After 4-5 days, the dead larvae are collected and macerated. The liquid is centrifuged and the pellet containing the viruses is suspended in sterile distilled water. This viral suspension can be used for spraying in the fields.

1Summary

Carbondioxide fixation and Biological Nitrogen Fixation are the most significant biological processes taking place on planet Earth. Methanogenesis is an anaerobic

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